is helpful to understand Angular ideas such as components, services, and modules.

React: Ionic also supports this framework if that's how you like to work. It's important to comprehend its component-based architecture.

- ***Node.js and npm/yarn***

Node.js: Since the Ionic CLI and other build tools rely on Node.js, it's critical to understand it.

npm/yarn: To manage dependencies and scripts, become familiar with package managers such as yarn or npm.

Setting up Ionic

Node.js is a JavaScript runtime that lets you run JavaScript on the server side. npm (node package manager) comes with Node.js and is used to manage JavaScript libraries and tools. The software requirements are:

- Operating system: Windows 10 or later, or macOS Mojave (10.14) or later (for iOS development), or Linux -- a recent version of your preferred Linux distribution
- Node.js and npm
- Ionic CLI
- Cordova
- A code editor (VS Code is recommended)
- Modern web browser: Chrome, Firefox, or Edge for testing and debugging

Go to the Node.js website and download the recommended LTS (long term support) version for your operating system.

Open the downloaded installer and follow the instructions to install Node.js and npm.

Next, verify the installation by running the following command in the terminal or command prompt depending on your operating system.

```
node -v
npm -v
```

This should display the versions of Node.js and npm, confirming that they are installed correctly.

To install the downloaded package on a Linux-based OS, we can follow the steps mentioned below:

```
sudo apt update
sudo apt install nodejs
sudo apt install npm
```

Check Node.js and npm versions:

```
node -v
npm -v
```

The Ionic command line interface (CLI) is used to create, build, and manage Ionic projects. Open the terminal and run:

```
npm install -g @ionic/cli
```

Verify the installation by typing:

```
ionic --version
```

Creating and installing Cordova

Cordova is a platform for building native mobile applications using web technologies like HTML, CSS, and JavaScript. To install it, run:

```
npm install -g cordova
```

The *-g* flag installs Cordova globally on your system, making the *cordova* command available from any directory.

Check the Cordova version to ensure it is installed correctly:

```
cordova -v
```

Creating a new Cordova project: Navigate to the directory where you want to create your project:

```
cd path/to/your/directory
cordova create myApp
```

myApp is the name of your new Cordova project. You can replace it with any name you prefer.

Navigate into your project directory and add platforms:

```
cd myApp
cordova platform add android
cordova platform add ios
```

To build the project for a specific platform, use the *build* command as follows:

```
cordova build android
cordova build ios
```

This will compile your project into an application for the selected platforms.

Creating an application using Ionic

Open the project in your *Code Editor*. You can use editors like Visual Studio Code. Edit *src/app/home/home.page.html*. Replace the contents of *home.page.html* with the following code:

```
<ion-header>
  <ion-toolbar>
    <ion-title>
      Hello World
    </ion-title>
  </ion-toolbar>
</ion-header>
<ion-content>
  <div style="text-align: center;
margin-top: 20px;">
    <h1>Hello World!</h1>
  </div>
</ion-content>
```

Open the terminal or command prompt and build the app:

```
ionic start myApp
```

You will be prompted to choose a framework.

Building and deploying the application

Open the terminal and build the app using the command given below: